

Worked examples for Lecture 15: Plane wave propagation

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These examples accompany Lecture 15. They are intended as practice in the concrete calculations that underlie the lecture, and are less demanding than the questions on the problem sets. Throughout, ρ is the referential density, A_{ijkl} the elastic tensor, $\Gamma(\hat{\mathbf{p}})$ the Christoffel matrix for a unit propagation direction $\hat{\mathbf{p}}$, and the phase speeds c and polarisation vectors $\mathbf{a}$ satisfy the Christoffel equation $\Gamma(\hat{\mathbf{p}})\mathbf{a} = c^2\mathbf{a}$. Numerical results were checked with the scripts in the python directory.

Example 1: P- and S-waves with numbers. At a depth of 100 km the Preliminary Reference Earth Model (PREM) has, to three significant figures, $\rho = 3.38 \times 10^3 \text{ kg m}^{-3}$, bulk modulus $\kappa = 1.30 \times 10^{11} \text{ Pa}$ and shear modulus $\mu = 6.80 \times 10^{10} \text{ Pa}$. Compute the Lamé parameter λ , the P- and S-wave speeds α and β , the ratio α/β , and Poisson's ratio $\sigma = \lambda/[2(\lambda + \mu)]$. Show that α/β depends on the elastic moduli only through σ , and check that $\alpha/\beta = \sqrt{3}$ for a Poisson solid, for which $\lambda = \mu$.

Solution. We recall from the lecture that the bulk modulus of an isotropic solid is $\kappa = \lambda + \frac{2}{3}\mu$, and hence

$$\lambda = \kappa - \frac{2}{3}\mu = 1.30 \times 10^{11} - 0.453 \times 10^{11} = 8.47 \times 10^{10} \text{ Pa.} \quad (1)$$

The P- and S-wave speeds follow from the solution of the Christoffel equation for an isotropic medium. It is convenient to write the P-wave modulus as $\lambda + 2\mu = \kappa + \frac{4}{3}\mu = 2.21 \times 10^{11} \text{ Pa}$, so that

$$\begin{aligned} \alpha &= \sqrt{\frac{\lambda + 2\mu}{\rho}} = \sqrt{\frac{2.207 \times 10^{11}}{3.38 \times 10^3}} \text{ m s}^{-1} = 8.08 \text{ km s}^{-1}, \\ \beta &= \sqrt{\frac{\mu}{\rho}} = \sqrt{\frac{6.80 \times 10^{10}}{3.38 \times 10^3}} \text{ m s}^{-1} = 4.49 \text{ km s}^{-1}. \end{aligned} \quad (2)$$

Note that with all quantities in SI units the speeds come out in m s^{-1} ; a common slip is to mix GPa with kg m^{-3} and obtain speeds that are wrong by a factor of $\sqrt{10^9}$. The ratio of the speeds and Poisson's ratio are

$$\frac{\alpha}{\beta} = \frac{8.080}{4.485} = 1.80, \quad \sigma = \frac{\lambda}{2(\lambda + \mu)} = \frac{8.47}{2 \times 15.27} = 0.277. \quad (3)$$

To relate the two, we note that the definition of σ can be inverted to give

$$\frac{\lambda}{\mu} = \frac{2\sigma}{1 - 2\sigma}, \quad (4)$$

and hence

$$\frac{\alpha^2}{\beta^2} = \frac{\lambda + 2\mu}{\mu} = 2 + \frac{\lambda}{\mu} = \frac{2(1 - \sigma)}{1 - 2\sigma}. \quad (5)$$

The ratio of the wave speeds is, therefore, a function of Poisson's ratio alone, and conversely a measurement of α/β determines σ without knowledge of the density. With $\sigma = 0.277$ eq. (5) gives $\alpha/\beta = 1.80$, in agreement with eq. (3). For a Poisson solid $\lambda = \mu$ gives $\sigma = 1/4$, and then $\alpha^2/\beta^2 = 2 \times \frac{3}{4}/\frac{1}{2} = 3$, so that $\alpha/\beta = \sqrt{3} = 1.73$.

The values in PREM are close to, but slightly above, those of a Poisson solid; a value of α/β between 1.7 and 1.8 is typical of the crust and mantle, which is why the Poisson solid is so often used as a rough model in seismology. It is worth remembering that in practice the logic runs the other way from this example: seismology measures α and β , and the moduli in PREM are inferred from them together with the density.

Example 2: The Christoffel matrix in a cubic crystal. A crystal with cubic symmetry has an elastic tensor of the form

$$A_{ijkl} = c_{12} \delta_{ij} \delta_{kl} + c_{44} (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) + (c_{11} - c_{12} - 2c_{44}) \sum_{n=1}^3 \delta_{in} \delta_{jn} \delta_{kn} \delta_{ln}, \quad (6)$$

where c_{11} , c_{12} and c_{44} are the three independent elastic constants in Voigt's notation and the co-ordinate axes are aligned with the crystal axes. Take the round values $c_{11} = 320$ GPa, $c_{12} = 70$ GPa, $c_{44} = 80$ GPa and $\rho = 3300$ kg m⁻³, which are of the magnitude found for olivine (though olivine itself is orthorhombic rather than cubic). Obtain the Christoffel matrix for a general unit vector $\hat{\mathbf{p}}$, and find the phase speeds and polarisation vectors for propagation along the [100], [110] and [111] directions, i.e. for $\hat{\mathbf{p}}$ proportional to (1,0,0), (1,1,0) and (1,1,1). For what relation between the constants is the material isotropic?

Solution. Writing $d = c_{11} - c_{12} - 2c_{44}$ for the coefficient of the last term in eq. (6), and contracting the elastic tensor with $\hat{p}_j \hat{p}_l$ term by term, we find

$$\rho \Gamma_{ik}(\hat{\mathbf{p}}) = A_{ijkl} \hat{p}_j \hat{p}_l = c_{12} \hat{p}_i \hat{p}_k + c_{44} (\delta_{ik} \hat{p}_j \hat{p}_j + \hat{p}_i \hat{p}_k) + d \sum_{n=1}^3 \delta_{in} \delta_{kn} \hat{p}_n^2, \quad (7)$$

where in the last term the sum over j and l has collapsed onto $j = l = n$. Since $\hat{p}_j \hat{p}_j = 1$, this can be written in matrix form as

$$\rho \Gamma(\hat{\mathbf{p}}) = (c_{12} + c_{44}) \hat{\mathbf{p}} \hat{\mathbf{p}}^T + c_{44} \mathbf{I} + d \text{diag}(\hat{p}_1^2, \hat{p}_2^2, \hat{p}_3^2). \quad (8)$$

The first two terms are exactly the isotropic Christoffel matrix of the lecture with $\lambda = c_{12}$ and $\mu = c_{44}$, and the third term, which is not invariant under rotations because it singles out the co-ordinate axes, carries all of the anisotropy. The material is therefore isotropic if and only if $d = 0$, i.e. $c_{11} - c_{12} = 2c_{44}$, in which case $c_{11} = \lambda + 2\mu$ as it

should. For our numbers $d = 320 - 70 - 160 = 90$ GPa, and the ratio $2c_{44}/(c_{11} - c_{12}) = 0.64$, which would be unity for an isotropic material, indicates a strongly anisotropic crystal.

Propagation along [100]. With $\hat{\mathbf{p}} = (1, 0, 0)$ the matrix $\hat{\mathbf{p}}\hat{\mathbf{p}}^T$ and the diagonal term are both non-zero only in the $(1, 1)$ entry, so that

$$\rho \Gamma = \text{diag}(c_{12} + 2c_{44} + d, c_{44}, c_{44}) = \text{diag}(c_{11}, c_{44}, c_{44}). \quad (9)$$

The eigenvectors are the co-ordinate axes: a longitudinal wave polarised along $\hat{\mathbf{e}}_1 = \hat{\mathbf{p}}$ with $\rho c^2 = c_{11}$, and a two-fold degenerate transverse wave with $\rho c^2 = c_{44}$ whose polarisation can be any vector in the $(\hat{\mathbf{e}}_2, \hat{\mathbf{e}}_3)$ -plane. Numerically $c = \sqrt{320 \times 10^9 / 3300} = 9.85 \text{ km s}^{-1}$ and $c = \sqrt{80 \times 10^9 / 3300} = 4.92 \text{ km s}^{-1}$.

Propagation along [110]. With $\hat{\mathbf{p}} = (1, 1, 0)/\sqrt{2}$ we have $\hat{p}_1^2 = \hat{p}_2^2 = \frac{1}{2}$, $\hat{p}_3 = 0$, and eq. (8) gives

$$\rho \Gamma = \frac{c_{12} + c_{44}}{2} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} + c_{44} \mathbf{I} + \frac{d}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (10)$$

Rather than expanding the characteristic polynomial we guess the eigenvectors from the structure of the matrix, as was done for the isotropic case in the lecture. The vector $(1, 1, 0)/\sqrt{2}$ is an eigenvector of the first matrix with eigenvalue $c_{12} + c_{44}$ and of the third with eigenvalue $d/2$; the vector $(1, -1, 0)/\sqrt{2}$ is annihilated by the first matrix and is an eigenvector of the third with eigenvalue $d/2$; and $(0, 0, 1)$ is annihilated by both. Hence

$$\begin{aligned} \mathbf{a} = \frac{1}{\sqrt{2}}(1, 1, 0) : \quad \rho c^2 &= c_{12} + 2c_{44} + \frac{1}{2}d = \frac{1}{2}(c_{11} + c_{12} + 2c_{44}) = 275 \text{ GPa}, \\ \mathbf{a} = \frac{1}{\sqrt{2}}(1, -1, 0) : \quad \rho c^2 &= c_{44} + \frac{1}{2}d = \frac{1}{2}(c_{11} - c_{12}) = 125 \text{ GPa}, \\ \mathbf{a} = (0, 0, 1) : \quad \rho c^2 &= c_{44} = 80 \text{ GPa}, \end{aligned} \quad (11)$$

giving phase speeds of 9.13, 6.15 and 4.92 km s⁻¹ respectively. The first wave is longitudinal and the other two transverse, but the two transverse waves now have different speeds: this is the shear wave splitting discussed at the end of the lecture, and here it is far from slight, the two S-wave speeds differing by some 25%. The polarisation of the faster S-wave lies in the $(1, 2)$ -plane, that of the slower one along the third crystal axis.

Propagation along [111]. With $\hat{\mathbf{p}} = (1, 1, 1)/\sqrt{3}$ all three $\hat{p}_n^2$ equal $\frac{1}{3}$, so that the anisotropic term is proportional to the identity and

$$\rho \Gamma = (c_{12} + c_{44}) \hat{\mathbf{p}}\hat{\mathbf{p}}^T + \left(c_{44} + \frac{1}{3}d\right) \mathbf{I}. \quad (12)$$

This has the same form as the isotropic Christoffel matrix, and so can be solved in the same way: $\hat{\mathbf{p}}$ is an eigenvector, and so is any vector orthogonal to it,

$$\begin{aligned} \mathbf{a} = \hat{\mathbf{p}} : \quad \rho c^2 &= c_{12} + 2c_{44} + \frac{1}{3}d = \frac{1}{3}(c_{11} + 2c_{12} + 4c_{44}) = 260 \text{ GPa}, \\ \mathbf{a} \perp \hat{\mathbf{p}} : \quad \rho c^2 &= c_{44} + \frac{1}{3}d = \frac{1}{3}(c_{11} - c_{12} + c_{44}) = 110 \text{ GPa}, \end{aligned} \quad (13)$$

with phase speeds 8.88 km s⁻¹ and (two-fold degenerate) 5.77 km s⁻¹. All of these eigenvalues were confirmed with a numerical eigen-decomposition of eq. (8).

Two features are worth noting. First, along the high-symmetry directions [100], [110] and [111] the waves are purely longitudinal or purely transverse; in a general direction the eigenvectors of eq. (8) are neither parallel nor perpendicular to $\hat{\mathbf{p}}$, which is why one speaks of quasi P- and quasi S-waves. Second, the S-wave degeneracy is not split along [111] even though the crystal is anisotropic: the [111] direction is a three-fold symmetry axis of the cube, and a symmetric 2×2 matrix acting in the plane transverse to such an axis that is invariant under rotations by 120° must be a multiple of the identity. Degeneracies in the slowness surface are thus often dictated by symmetry.

Example 3: The SH sheet of the slowness surface in a transversely isotropic medium. A transversely isotropic medium with symmetry axis $\hat{\mathbf{v}}$ has elastic tensor

$$A_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{kj}) + 8\gamma \hat{v}_i \hat{v}_j \hat{v}_k \hat{v}_l + 4\zeta (\hat{v}_i \hat{v}_j \delta_{kl} + \delta_{ij} \hat{v}_k \hat{v}_l) - \zeta (\hat{v}_i \hat{v}_k \delta_{jl} + \hat{v}_j \hat{v}_k \delta_{il} + \hat{v}_j \hat{v}_l \delta_{ik} + \hat{v}_i \hat{v}_l \delta_{jk}), \quad (14)$$

as in Lecture 13 and the first problem set. Take $\hat{\mathbf{v}} = \hat{\mathbf{e}}_3$ and $\hat{\mathbf{p}} = (\sin \theta, 0, \cos \theta)$. Show that $\hat{\mathbf{e}}_2$ is an exact eigenvector of $\Gamma(\hat{\mathbf{p}})$ for every θ , and find the corresponding phase speed. Show that the associated sheet of the slowness surface is an ellipsoid of revolution, and compare the exact phase speed with the prediction of first-order degenerate perturbation theory. Illustrate the result for the sample values $\rho = 3300 \text{ kg m}^{-3}$, $\lambda = 80$, $\mu = 70$, $\gamma = 2$, $\zeta = 1$ and $\xi = 8 \text{ GPa}$.

Solution. We first form the Christoffel matrix for a general direction. Contracting each term of eq. (14) with $\hat{p}_j \hat{p}_l$, and writing $\hat{\mathbf{v}} \cdot \hat{\mathbf{p}} = \cos \theta$, we obtain

$$\rho \Gamma_{ik}(\hat{\mathbf{p}}) = (\lambda + \mu) \hat{p}_i \hat{p}_k + \mu \delta_{ik} + 8\gamma \cos^2 \theta \hat{v}_i \hat{v}_k + 4\zeta \cos \theta (\hat{v}_i \hat{p}_k + \hat{p}_i \hat{v}_k) - \zeta [\hat{v}_i \hat{v}_k + \cos \theta (\hat{v}_i \hat{p}_k + \hat{p}_i \hat{v}_k) + \cos^2 \theta \delta_{ik}]. \quad (15)$$

Now let this matrix act on $\hat{\mathbf{e}}_2$. Every term in eq. (15) other than the two proportional to δ_{ik} contains a factor of either $\hat{p}_k$ or $\hat{v}_k$, and both $\hat{p}_k \hat{e}_{2k} = 0$ and $\hat{v}_k \hat{e}_{2k} = 0$ because $\hat{\mathbf{e}}_2$ is orthogonal to the plane containing the propagation direction and the symmetry axis. Hence

$$\Gamma(\hat{\mathbf{p}}) \hat{\mathbf{e}}_2 = \frac{\mu - \zeta \cos^2 \theta}{\rho} \hat{\mathbf{e}}_2, \quad (16)$$

so that $\hat{\mathbf{e}}_2$ is an exact eigenvector for all θ , with squared phase speed

$$c_{\text{SH}}^2 = \frac{\mu - \zeta \cos^2 \theta}{\rho} = \frac{\mu \sin^2 \theta + (\mu - \zeta) \cos^2 \theta}{\rho}. \quad (17)$$

This wave is transverse and polarised horizontally (if we think of $\hat{\mathbf{v}}$ as vertical), and so is called the SH wave. Its speed varies from $\sqrt{\mu/\rho}$ for horizontal propagation to $\sqrt{(\mu - \zeta)/\rho}$ along the axis, and a positive squared speed along the axis requires $\mu > \zeta$. Because eigenvectors of a symmetric matrix with distinct eigenvalues are orthogonal, the remaining two polarisation vectors lie in the $(\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_3)$ -plane spanned by $\hat{\mathbf{p}}$ and $\hat{\mathbf{v}}$; they are the quasi P- and quasi SV-waves, whose speeds require the solution of a genuine 2×2 eigenvalue problem and are not in general given by such a simple formula. Nothing in the argument depended on $\hat{\mathbf{p}}$ lying in the $(1, 3)$ -plane: by the rotational

symmetry about $\hat{\nu}$, for any $\hat{\mathbf{p}}$ making an angle θ with the axis the vector $\hat{\nu} \times \hat{\mathbf{p}} / \sin \theta$ is an exact eigenvector with the eigenvalue in eq. (17).

The slowness sheet. The slowness vector of the SH wave is $\mathbf{p} = \hat{\mathbf{p}} / c_{\text{SH}}$, so that $\|\mathbf{p}\|^2 = \rho / (\mu - \zeta \cos^2 \theta)$ and $p_3 = \|\mathbf{p}\| \cos \theta$. Multiplying out,

$$\mu \|\mathbf{p}\|^2 - \zeta p_3^2 = \rho \iff \mu(p_1^2 + p_2^2) + (\mu - \zeta)p_3^2 = \rho, \quad (18)$$

which is an ellipsoid of revolution about the symmetry axis with equatorial semi-axis $\sqrt{\rho/\mu}$ and polar semi-axis $\sqrt{\rho/(\mu - \zeta)}$. For $\zeta > 0$ the sheet is prolate, the slowness being largest, and the speed smallest, along the axis. That one sheet of the slowness surface of a transversely isotropic medium is exactly an ellipsoid is a well-known special property; the quasi-P and quasi-SV sheets are quartic surfaces.

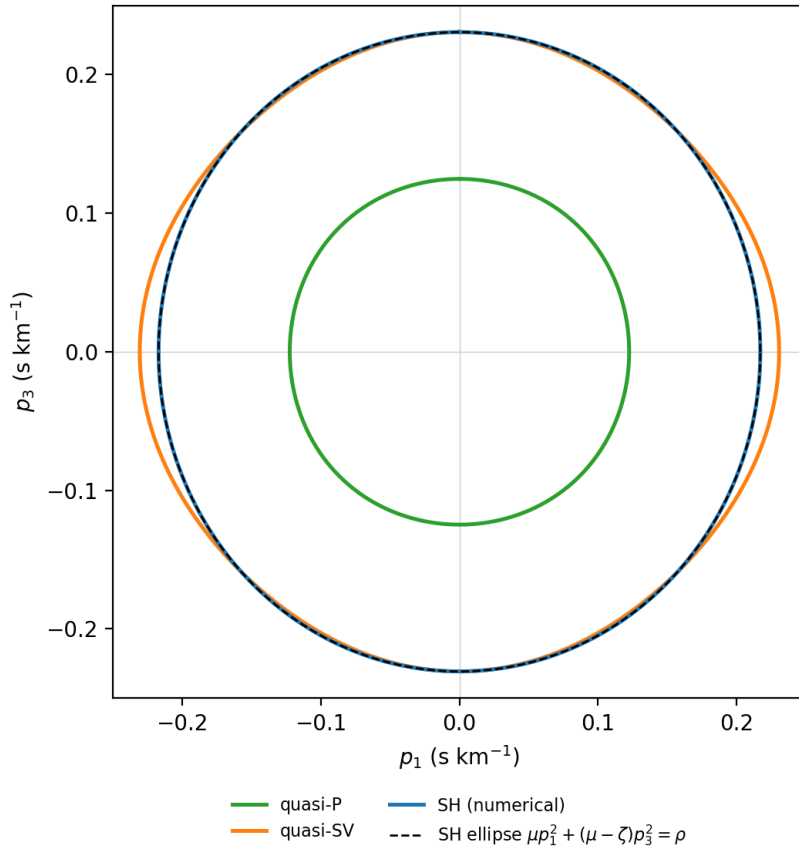


Fig. 1: Section through the slowness surface in the (p_1, p_3) -plane for the sample transversely isotropic medium of Example 3, with the symmetry axis vertical. The three sheets were obtained by numerical diagonalisation of the Christoffel matrix and are labelled by their polarisation; the dashed curve is the ellipse of eq. (18), on which the numerical SH sheet lies to machine precision. The SH and quasi-SV sheets touch on the symmetry axis, where the two S-waves are degenerate, and cross at oblique angles.

Comparison with perturbation theory. Treating the terms in γ , $\tilde{\zeta}$ and ζ as the perturbation $\delta\Gamma$, with the isotropic part having $\beta^2 = \mu/\rho$, the degenerate perturbation theory of the lecture asks us to diagonalise $\hat{\mathbf{t}}_a \cdot \delta\Gamma(\hat{\mathbf{p}})\hat{\mathbf{t}}_b$ on the plane orthogonal to $\hat{\mathbf{p}}$. Taking $\hat{\mathbf{t}}_2 =$

$\hat{\mathbf{e}}_2$ and $\hat{\mathbf{t}}_1 = \hat{\mathbf{e}}_2 \times \hat{\mathbf{p}} = (\cos \theta, 0, -\sin \theta)$, eq. (16) shows that $\delta \Gamma \hat{\mathbf{e}}_2 = -(\zeta \cos^2 \theta / \rho) \hat{\mathbf{e}}_2$, and hence

$$\hat{\mathbf{t}}_1 \cdot \delta \Gamma(\hat{\mathbf{p}}) \hat{\mathbf{t}}_2 = 0, \quad \hat{\mathbf{t}}_2 \cdot \delta \Gamma(\hat{\mathbf{p}}) \hat{\mathbf{t}}_2 = -\frac{\zeta \cos^2 \theta}{\rho}. \quad (19)$$

The 2×2 matrix is diagonal in this basis, so $\hat{\mathbf{e}}_2$ is one of the zeroth-order polarisations (as it must be, being an exact eigenvector), and the first-order speed perturbation of the SH wave is

$$\delta \beta_{\text{SH}} = \frac{1}{2\beta} \hat{\mathbf{e}}_2 \cdot \delta \Gamma(\hat{\mathbf{p}}) \hat{\mathbf{e}}_2 = -\frac{\zeta \cos^2 \theta}{2\rho\beta} = -\frac{\zeta}{2\mu} \beta \cos^2 \theta. \quad (20)$$

The exact result, eq. (17), can be written

$$c_{\text{SH}} = \beta \sqrt{1 - \frac{\zeta}{\mu} \cos^2 \theta} = \beta \left[1 - \frac{\zeta}{2\mu} \cos^2 \theta - \frac{\zeta^2}{8\mu^2} \cos^4 \theta - \dots \right], \quad (21)$$

which agrees with $\beta + \delta \beta_{\text{SH}}$ to first order in ζ/μ , the error being of second order as it should be. The perturbation theory gives the eigenvalue c^2 exactly here; the only approximation is the linearisation of the square root.

For the sample values the SH speed is 4.61 km s^{-1} horizontally and 4.33 km s^{-1} along the axis, and with $\zeta/\mu = 0.114$ the first-order formula is in error by at most 8 m s^{-1} , or 0.2%. Fig. 1 shows the section of the slowness surface through the (p_1, p_3) -plane. Along the axis ($\theta = 0$) the SH and quasi-SV sheets touch, since eq. (15) shows that $\Gamma(\hat{\nu})$ acts as $(\mu - \zeta)/\rho$ times the identity on the plane orthogonal to $\hat{\nu}$: the two S-waves are degenerate along a symmetry axis, exactly as along $[111]$ in the cubic crystal.

Example 4: First-order perturbation theory checked against exact results. Write the elastic tensor of the cubic crystal of Example 2 as $A_{ijkl} = c_{12}\delta_{ij}\delta_{kl} + c_{44}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) + s\delta A_{ijkl}$ with

$$\delta A_{ijkl} = d \sum_{n=1}^3 \delta_{in}\delta_{jn}\delta_{kn}\delta_{ln}, \quad d = c_{11} - c_{12} - 2c_{44}, \quad (22)$$

so that $s = 0$ is an isotropic medium with $\lambda = c_{12}$, $\mu = c_{44}$, and $s = 1$ is the crystal itself. (a) Obtain $\delta \Gamma(\hat{\mathbf{p}})$ and the first-order quasi-P speed perturbation $\delta \alpha$ for a general direction. (b) For $\hat{\mathbf{p}}$ along $[110]$ compare $\alpha + \delta \alpha$ with the exact speed found in Example 2, and account for the discrepancy. (c) Solve the 2×2 degenerate problem for the quasi S-waves along $[110]$ and show that it reproduces the exact splitting between $\rho c^2 = \frac{1}{2}(c_{11} - c_{12})$ and $\rho c^2 = c_{44}$. (d) For the general direction $\hat{\mathbf{p}} = (2, 3, 6)/7$ compare the three first-order eigenvalues with the exact ones, computed numerically, for $s = 1$, $\frac{1}{2}$ and $\frac{1}{4}$, and hence exhibit the order of the error.

Solution. (a) The unperturbed medium has $\alpha^2 = (c_{12} + 2c_{44})/\rho$ and $\beta^2 = c_{44}/\rho$, which for the numbers of Example 2 give $\alpha = 8.35 \text{ km s}^{-1}$ and $\beta = 4.92 \text{ km s}^{-1}$. From eq. (8), or directly from the definition $\delta \Gamma_{ik} = \rho^{-1} \delta A_{ijkl} \hat{p}_j \hat{p}_l$,

$$\rho \delta \Gamma(\hat{\mathbf{p}}) = d \text{diag}(\hat{p}_1^2, \hat{p}_2^2, \hat{p}_3^2). \quad (23)$$

The non-degenerate first-order formula of the lecture then gives, for the quasi P-wave,

$$\delta\alpha = \frac{1}{2\alpha} \hat{\mathbf{p}} \cdot \delta\mathbf{\Gamma}(\hat{\mathbf{p}}) \hat{\mathbf{p}} = \frac{d}{2\rho\alpha} (\hat{p}_1^4 + \hat{p}_2^4 + \hat{p}_3^4). \quad (24)$$

Since $\sum_n \hat{p}_n^4$ ranges from $\frac{1}{3}$ (along [111]) to 1 (along a cube axis), the quasi-P speed varies by an amount of order $d/(3\rho\alpha)$ with direction; for $d > 0$ the fastest direction is a cube axis and the slowest is a body diagonal, in agreement with the exact results of Example 2.

(b) Along [110] we have $\hat{p}_1^2 = \hat{p}_2^2 = \frac{1}{2}$ and $\hat{p}_3 = 0$, so that $\sum_n \hat{p}_n^4 = \frac{1}{2}$ and

$$\delta\alpha = \frac{d}{4\rho\alpha} = \frac{90 \times 10^9}{4 \times 3300 \times 8348} \text{ m s}^{-1} = 0.817 \text{ km s}^{-1}, \quad \alpha + s \delta\alpha|_{s=1} = 9.17 \text{ km s}^{-1}. \quad (25)$$

The exact speed from Example 2 is $\sqrt{275 \times 10^9 / 3300} = 9.13 \text{ km s}^{-1}$, so the first-order result is too large by 37 m s^{-1} , or 0.4%. To see where this comes from, note that along [110] the matrix $\rho \delta\mathbf{\Gamma} = \frac{1}{2}d \text{ diag}(1, 1, 0)$ maps $\hat{\mathbf{p}}$ to $\frac{1}{2}d \hat{\mathbf{p}}$, so $\hat{\mathbf{p}}$ is an exact eigenvector of the full Christoffel matrix, with the exact eigenvalue

$$c^2 = \alpha^2 + s \frac{d}{2\rho} = \alpha^2 + 2\alpha s \delta\alpha. \quad (26)$$

First-order perturbation theory is therefore exact for the eigenvalue c^2 ; the entire discrepancy arises from the expansion of the square root,

$$c = \alpha \sqrt{1 + \frac{2s \delta\alpha}{\alpha}} = \alpha + s \delta\alpha - \frac{s^2 \delta\alpha^2}{2\alpha} + \dots, \quad (27)$$

and the neglected second-order term, $-\delta\alpha^2/(2\alpha) = -d^2/(32\rho^2\alpha^3) = -40 \text{ m s}^{-1}$ at $s = 1$, accounts for the difference to within the size of the third-order term. The expansion parameter here is $2\delta\alpha/\alpha = d/(2\rho\alpha^2) = 90/460 = 0.20$, so an error of order its square is to be expected.

(c) For the quasi S-waves along [110] we take the orthonormal transverse vectors $\hat{\mathbf{t}}_1 = (0, 0, 1)$ and $\hat{\mathbf{t}}_2 = (1, -1, 0)/\sqrt{2}$. With $\rho \delta\mathbf{\Gamma} = \frac{1}{2}d \text{ diag}(1, 1, 0)$ we find $\hat{\mathbf{t}}_1 \cdot \delta\mathbf{\Gamma} \hat{\mathbf{t}}_1 = 0$, since $\hat{\mathbf{t}}_1$ has no component in the (1,2)-plane, $\hat{\mathbf{t}}_2 \cdot \delta\mathbf{\Gamma} \hat{\mathbf{t}}_2 = d/(2\rho)$, and $\hat{\mathbf{t}}_1 \cdot \delta\mathbf{\Gamma} \hat{\mathbf{t}}_2 = 0$. The degenerate eigenvalue problem of the lecture is thus already diagonal,

$$\begin{pmatrix} 0 & 0 \\ 0 & d/(2\rho) \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 2\beta \delta\beta \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}, \quad (28)$$

with solutions $\delta\beta = 0$ for polarisation $\hat{\mathbf{t}}_1 = (0, 0, 1)$ and $\delta\beta = d/(4\rho\beta)$ for polarisation $\hat{\mathbf{t}}_2 = (1, -1, 0)/\sqrt{2}$. These are precisely the exact polarisations found in Example 2, and the corresponding eigenvalues are

$$c^2 = \beta^2 = \frac{c_{44}}{\rho}, \quad c^2 = \beta^2 + 2\beta \delta\beta = \frac{c_{44} + \frac{1}{2}d}{\rho} = \frac{c_{11} - c_{12}}{2\rho}, \quad (29)$$

which are the exact values. The reason is the same as in part (b): because $\delta\mathbf{\Gamma}$ is diagonal in the basis $\{\hat{\mathbf{p}}, \hat{\mathbf{t}}_1, \hat{\mathbf{t}}_2\}$, these three vectors are exact eigenvectors of $\mathbf{\Gamma} + s \delta\mathbf{\Gamma}$ for every s , and the eigenvalues are exactly linear in s . The first-order theory correctly predicts

that the S-wave degeneracy is split, and which polarisation is the faster. For the speeds themselves, however, the linearised result $\beta + \delta\beta = 4.92 + 1.38 = 6.31 \text{ km s}^{-1}$ compares poorly with the exact $\sqrt{125 \times 10^9 / 3300} = 6.15 \text{ km s}^{-1}$, the error of 2.5% again coming from the square root; the expansion parameter is now $d/(2\rho\beta^2) = 45/80 = 0.56$, and the anisotropy of this crystal is simply not slight as far as its S-waves are concerned.

(d) In a general direction none of the zeroth-order vectors is an exact eigenvector and the first-order eigenvalues are genuinely approximate. For $\hat{\mathbf{p}} = (2, 3, 6)/7$ we have $\sum_n \hat{p}_n^4 = 1393/2401 = 0.580$, so the first-order quasi-P eigenvalue is

$$\rho c^2 = c_{12} + 2c_{44} + s d \sum_n \hat{p}_n^4 = (230 + 52.22 s) \text{ GPa}. \quad (30)$$

For the quasi S-waves we choose $\hat{\mathbf{t}}_1 \propto \hat{\mathbf{p}} \times \hat{\mathbf{e}}_3$ and $\hat{\mathbf{t}}_2 = \hat{\mathbf{p}} \times \hat{\mathbf{t}}_1$, and evaluate the 2×2 matrix numerically from eq. (23),

$$\rho (\hat{\mathbf{t}}_a \cdot \delta \Gamma \hat{\mathbf{t}}_b) = \begin{pmatrix} 10.17 & -3.63 \\ -3.63 & 27.61 \end{pmatrix} \text{ GPa}, \quad (31)$$

whose eigenvalues are 9.45 and 28.34 GPa (their sum, 37.78 GPa, equals $d(1 - \sum_n \hat{p}_n^4)$, as it must since the trace of $\rho \delta \Gamma$ is d). The first-order quasi-S eigenvalues are therefore $\rho c^2 = (80 + 9.45 s)$ and $(80 + 28.34 s)$ GPa. Comparing with the exact eigenvalues of $\Gamma + s \delta \Gamma$ obtained numerically, all in GPa, we find:

s	quasi S ₁			quasi S ₂			quasi P		
	first order	exact	error	first order	exact	error	first order	exact	error
1	89.45	89.18	+0.27	108.34	105.58	+2.76	282.22	285.25	-3.03
1/2	84.72	84.65	+0.07	94.17	93.41	+0.76	256.11	256.93	-0.83
1/4	82.36	82.34	+0.02	87.08	86.89	+0.20	243.05	243.27	-0.22

Halving s divides each error by a factor close to four (between 3.8 and 3.9, the shortfall being due to third-order terms), confirming that the first-order eigenvalues are in error at order s^2 . Two further features of the table are instructive. The three errors in each row sum to zero, because the trace of $\Gamma + s \delta \Gamma$ is exactly linear in s and is reproduced by first-order theory. And the quasi-P eigenvalue is underestimated while both quasi-S eigenvalues are overestimated: this is the familiar “level repulsion” of second-order perturbation theory, the second-order shift in c^2 of the quasi P-wave being

$$s^2 \sum_{a=1}^2 \frac{[\hat{\mathbf{t}}_a \cdot \delta \Gamma(\hat{\mathbf{p}}) \hat{\mathbf{p}}]^2}{\alpha^2 - \beta^2} = 3.60 s^2 \text{ GPa} / \rho, \quad (32)$$

which is positive because the P-wave has the largest eigenvalue, and which reproduces the quasi-P errors in the table for the smaller values of s (at $s = \frac{1}{4}$ it gives 0.225 GPa against the observed 0.22 GPa). Translating to speeds at $s = 1$, the lecture formulae give $\alpha + \delta\alpha = 9.30$, $\beta + \delta\beta_1 = 5.21$ and $\beta + \delta\beta_2 = 5.80 \text{ km s}^{-1}$, against exact values of 9.30, 5.20 and 5.66 km s^{-1} . The remarkable accuracy of the quasi-P speed here is partly fortuitous: the neglected second-order shift in c^2 (positive) and the neglected term in the square-root expansion (negative) happen almost to cancel in this direction, whereas for the faster quasi S-wave both errors have the same sign and add.

The moral is that first-order perturbation theory captures the qualitative picture – the directional dependence of the quasi-P speed, the splitting of the S-waves and

their polarisations – exactly as the lecture describes, while its quantitative accuracy is governed by the ratio of the anisotropic perturbation to the relevant isotropic modulus. For the weak anisotropy of a few per cent typical of the Earth's mantle this ratio is small and the theory is excellent; for a single olivine-like crystal it is not.

Example 5: Energy flux of a plane wave. On the first problem set it is shown that, in the absence of a stress glut, the energy density $E = \frac{1}{2}\rho \partial_t u_i \partial_t u_i + \frac{1}{2}A_{ijkl} \partial_j u_i \partial_l u_k$ satisfies $\partial_t E + \partial_j s_j = 0$, where the elastic Poynting vector is $s_j = -A_{ijkl} \partial_l u_k \partial_t u_i$. Consider the plane wave $u_i(\mathbf{x}, t) = f(t - p_j x_j) a_i$ with $\|\mathbf{a}\| = 1$ satisfying the Christoffel equation. Show that the energy flux is along the vector $A_{ijkl} a_i a_k p_l$, and that the **energy velocity** $\mathbf{v} = \mathbf{s}/E$ satisfies $\mathbf{v} \cdot \mathbf{p} = 1$. Show that for a P-wave in an isotropic medium the time-averaged flux is $\langle \mathbf{s} \rangle = \rho \alpha \langle \dot{u}^2 \rangle \hat{\mathbf{p}}$, with the corresponding result for an S-wave, and find the energy velocity of the SH wave of Example 3.

Solution. For the plane wave the derivatives were computed in the lecture,

$$\frac{\partial u_k}{\partial x_l} = -f'(t - p_m x_m) a_k p_l, \quad \frac{\partial u_i}{\partial t} = f'(t - p_m x_m) a_i, \quad (33)$$

and so, on substituting into the definitions,

$$s_j = f'^2 A_{ijkl} a_i a_k p_l, \quad E = \frac{1}{2} f'^2 (\rho a_i a_i + A_{ijkl} a_i p_j a_k p_l) = \frac{1}{2} f'^2 \rho (1 + \mathbf{a} \cdot \boldsymbol{\Gamma}(\mathbf{p}) \mathbf{a}) = \rho f'^2, \quad (34)$$

where in the last step we used the Christoffel equation $\boldsymbol{\Gamma}(\mathbf{p}) \mathbf{a} = \mathbf{a}$ in its unnormalised form (recall $\boldsymbol{\Gamma}(\mathbf{p}) = c^{-2} \boldsymbol{\Gamma}(\hat{\mathbf{p}})$) together with $\|\mathbf{a}\| = 1$. Thus the kinetic and strain energies of a plane wave are equal at every instant, and the energy flux at every instant is along the fixed vector $A_{ijkl} a_i a_k p_l$; the same is therefore true of the time average, which for a periodic f is $\langle s_j \rangle = \langle f'^2 \rangle A_{ijkl} a_i a_k p_l$. The energy velocity is independent of f ,

$$v_j = \frac{s_j}{E} = \frac{1}{\rho} A_{ijkl} a_i a_k p_l, \quad \mathbf{v} \cdot \mathbf{p} = \frac{1}{\rho} A_{ijkl} a_i p_j a_k p_l = \mathbf{a} \cdot \boldsymbol{\Gamma}(\mathbf{p}) \mathbf{a} = 1. \quad (35)$$

Since $\mathbf{p} = \hat{\mathbf{p}}/c$, this says that $\mathbf{v} \cdot \hat{\mathbf{p}} = c$: the component of the energy velocity along the propagation direction is always the phase speed, but in an anisotropic medium $\mathbf{v}$ need not be parallel to $\hat{\mathbf{p}}$, in which case $\|\mathbf{v}\| > c$ and energy travels obliquely to the wavefronts. This energy velocity is what is called the group velocity in the study of dispersive or anisotropic waves.

Isotropic P- and S-waves. Inserting the isotropic elastic tensor,

$$A_{ijkl} a_i a_k p_l = \lambda a_j (\mathbf{a} \cdot \mathbf{p}) + \mu (\mathbf{a} \cdot \mathbf{a}) p_j + \mu a_j (\mathbf{a} \cdot \mathbf{p}). \quad (36)$$

For a P-wave $\mathbf{a} = \hat{\mathbf{p}}$ and $\mathbf{p} = \hat{\mathbf{p}}/\alpha$, so this becomes $(\lambda + 2\mu)\hat{\mathbf{p}}/\alpha = \rho \alpha \hat{\mathbf{p}}$, while for an S-wave $\mathbf{a} \cdot \mathbf{p} = 0$ and $\mathbf{p} = \hat{\mathbf{p}}/\beta$ give $\mu \hat{\mathbf{p}}/\beta = \rho \beta \hat{\mathbf{p}}$. Since the particle velocity is $\dot{u}_i = f' a_i$ with $\|\mathbf{a}\| = 1$, we have $\dot{u}^2 = f'^2$, and hence

$$\langle \mathbf{s} \rangle = \rho \alpha \langle \dot{u}^2 \rangle \hat{\mathbf{p}} \quad (\text{P-wave}), \quad \langle \mathbf{s} \rangle = \rho \beta \langle \dot{u}^2 \rangle \hat{\mathbf{p}} \quad (\text{S-wave}). \quad (37)$$

In an isotropic medium, then, energy is carried along the propagation direction at the phase speed, with a flux equal to the wave speed times the energy density $\rho \langle \dot{u}^2 \rangle$, and

the product $\rho\alpha$ or $\rho\beta$ that appears is the acoustic impedance familiar from the theory of reflection and transmission.

The SH wave in a transversely isotropic medium. With $\mathbf{a} = \hat{\mathbf{e}}_2$ and $\hat{\mathbf{v}} = \hat{\mathbf{e}}_3$ we need $A_{2j2l}p_l$. From eq. (14), and using $\hat{v}_2 = 0$, the only surviving terms are $A_{2j2l} = \lambda\delta_{2j}\delta_{2l} + \mu(\delta_{jl} + \delta_{2j}\delta_{2l}) - \zeta\hat{v}_j\hat{v}_l$, and since $p_2 = 0$,

$$\mathbf{v} = \frac{1}{\rho} (\mu\mathbf{p} - \zeta(\hat{\mathbf{v}} \cdot \mathbf{p})\hat{\mathbf{v}}) = \frac{1}{\rho c_{\text{SH}}} (\mu \sin \theta, 0, (\mu - \zeta) \cos \theta). \quad (38)$$

This is parallel to $\hat{\mathbf{p}} = (\sin \theta, 0, \cos \theta)$ only for $\theta = 0$ or $\pi/2$; in between, the energy velocity is deflected away from the symmetry axis (for $\zeta > 0$), by an angle whose maximum for the sample values of Example 3 is about 3.5° , at $\theta \approx 43^\circ$. One can check directly from eq. (17) that

$$\mathbf{v} \cdot \hat{\mathbf{p}} = \frac{\mu \sin^2 \theta + (\mu - \zeta) \cos^2 \theta}{\rho c_{\text{SH}}} = c_{\text{SH}}, \quad (39)$$

as required by eq. (35). Finally, the gradient with respect to $\mathbf{p}$ of the function $\mu(p_1^2 + p_2^2) + (\mu - \zeta)p_3^2 - \rho$ defining the SH slowness sheet in eq. (18) is $2(\mu p_1, \mu p_2, (\mu - \zeta)p_3)$, which is parallel to $\mathbf{v}$: the energy velocity is normal to the slowness surface. This is in fact a general property of plane waves in anisotropic media, and it is the reason that the shape of the slowness surface, rather than the phase speed alone, governs the geometry of wave propagation.